

Assignment Operators

Efficiently Updating and Managing Data

Assignment operators are used to store or update values in variables. While the `=` is the most basic, Java provides "Compound" shortcuts to help you write cleaner code.

The Shortcut Rule:

`x += 5` is simply a shorter way of writing `x = x + 5`.

1. The Common Shortcuts

Operator	Example	Matches This Logic
<code>=</code>	<code>x = 5</code>	<code>x = 5</code>
<code>+=</code>	<code>x += 3</code>	<code>x = x + 3</code>
<code>-=</code>	<code>x -= 3</code>	<code>x = x - 3</code>
<code>*=</code>	<code>x *= 3</code>	<code>x = x * 3</code>
<code>/=</code>	<code>x /= 3</code>	<code>x = x / 3</code>
<code>%=</code>	<code>x %= 3</code>	<code>x = x % 3</code>

2. Advanced (Bitwise) Shortcuts

These are used for managing bits and binary data—useful once you reach systems programming or complex flags.

Operator	Behavior
<code>&=</code> , <code> =</code> , <code>^=</code>	AND, OR, and XOR assignment
<code>>>=</code> , <code><<=</code>	Shift assignment (Right and Left)

3. Real-Life Logic: Savings Tracker

Imagine you are building a banking app. Use `+=` to add deposits to a balance.

```
int savings = 100;
// Add $50 deposit
savings += 50;

System.out.println("Total balance: $" + savings);
// Output: Total balance: $150
```

Key takeaways

- **Assign:** Use `=` for the first time you set a variable.
- **Update:** Use `+=`, `-=`, etc., to update existing values.
- **Efficiency:** These shortcuts make your code faster to type and easier for other developers to read.
- **Counters:** They are most commonly used for scoreboards, loops, and accumulating totals.

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